

AI can predict heart disease – but can we trust how it makes decisions?

New research using explainable machine learning reveals that understanding AI decisions in healthcare might be more complicated than it appears

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AI is transforming healthcare, with a catch

Artificial Intelligence is increasingly being used in modern healthcare. From analysing images to predicting disease risk, machine learning systems are increasingly being used to support clinical decision making. In the case of cardiovascular disease which is one of the leading causes of death globally these tools allow earlier detection and more targeted interventions, hopefully saving lives and reducing pressure on healthcare systems.

This growing ability comes with a problem. Many of the most accurate machine learning models are “black boxes”, giving predictions without explaining how decisions were made. A model can identify a patient as high-risk for heart disease, but show no reasoning that can be interpreted, questioned or validated. In a domain where these decisions affect patient outcomes, this isn't only a limitation but a barrier.

To address this, researchers have tried using explainable artificial intelligence (XAI), a set of techniques designed to make machine learning more understandable. These methods try to provide insight into how models make their predictions, bridging technical performance and usability. However, whether these explanations improve understanding or create more challenges is still a question.

This creates a challenge, in healthcare, it isn't enough for the systems to be accurate, it also needs to be interpretable, especially when decisions affect treatment. As a result, there is interest in techniques that make complex models understandable without reducing performance.

To explore this, I built a heart disease prediction system that applies two widely used explanation techniques, SHAP and LIME, to examine if AI explanations are consistent and meaningful in practice

Why explainability matters for patients and society

The importance of explainability in healthcare AI is far beyond technical curiosity, it shapes how these systems are trusted, used and integrated into real world decision making. An AI system that can't explain its reasoning disrupts this process, making it difficult to assess if results are reliable or appropriate for individual patients.

However, the issue is not only about trust, but also about fairness and inclusion. Many machine learning models are trained on publicly available datasets that might not represent the diversity of patients. If certain demographics are underrepresented or misrepresented, predictions would be less accurate for those individuals, reinforcing healthcare inequalities. Without clear explanations these biases can remain hidden making them harder to challenge.

There is also risk of over-reliance. Research has found that people place confidence in automated systems, especially when outputs are authoritative or are presented with

supportive explanations. In healthcare, this can lead to automation bias, where clinicians defer to AI recommendations even when they conflict with their own judgement. Poorly designed or misleading explanations reinforce this effect, creating a false sense of confidence rather than improving understanding.

The legal challenge: who is responsible

As artificial intelligence becomes more embedded in healthcare, legal responsibilities are harder to ignore. When an AI contributes to a clinical decision especially one that leads to harm, it's not clear who should be held accountable. Is responsibility held by the clinician who relied on the system, the developer who made it, or the organisation who deployed it?

One of the most frequently cited legal protections in this area is the General Data Protection Regulation (GDPR) which governs how personal data is processed and used. In the context of automated decision making, GDPR is interpreted as providing individuals with a 'right to explanation'. However, this right is more limited than most people think. The regulation requires that individuals are given a meaningful information about how decisions are made, but does not specify what form that should take, it doesn't guarantee a fully transparent account of complex model behaviour.

More recently, efforts have begun to address these gaps. The proposed EU artificial intelligence act would classify many healthcare AI systems as 'high risk' placing strict requirements on transparency, accountability, and risk management. Similarly, guidance from UK healthcare emphasises the need for explainability as a part of safe and trustworthy AI deployment.

However, regulation still struggles to keep pace with technology complexity. While laws can mandate transparency in principle, they don't resolve the question of what makes a good explanation in a clinical context. If different methods produce conflicting interpretations of the same prediction, it becomes unclear which one satisfies legal requirements. In this sense, the challenge is not only about enforcing transparency but about defining what meaningful transparency looks like in practice.

My own findings show this directly. When SHAP and LIME were applied to the same prediction, they sometimes ranked features differently. If a clinician was shown either explanation as the basis of the rejection, they would get a different account of the same output. Current regulation offers no answers to which explanation is legally sufficient or who is responsible when they conflict.

The ethical problem: can explanations be trusted?

While explainable AI is often presented as a solution to the opacity of machine learning, it creates a subtle ethical challenge: can the explanations themselves be trusted? If explanations are inconsistent, unstable or misleading, they risk creating an illusion instead of understanding.

Different explanation methods can produce meaningfully different interpretations of the same model. In my project, two techniques (SHAP and LIME) were applied to the same predictions, and they didn't always agree on the feature importance. At a global level, where features are averaged across many patients, the methods suggest similar patterns. However, at the level of individual predictions, where decisions are made, these differences be more pronounced.

There are also concerns about reliability. Some techniques, especially those based on local approximations, can produce different results for the same input. This instability is a problem in healthcare since consistency is essential for trust. An explanation that changes depending on how it is generated lowers the confidence in both model and the explanation.

Even when explanations are stable, they can be misleading. Methods like SHAP assign importance values to features based on statistical contribution, but they don't necessarily show causal relationships. A feature can appear influential in a prediction without a direct cause of the outcome. Without interpretation, users can draw incorrect assumptions about why a decision was made, especially when outputs are simplified for non-technical audiences.

Making sure that the explanations are accurate, consistent and meaningful is therefore not a technical issue but also an ethical one.

Can we make AI understandable?

The challenge of making AI understandable to people who use it is harder than it seems. Building a system that produces an explanation is straightforward. But building one that is also accurate, meaningful and interpreted correctly by non-technical users is a whole different problem.

While developing my dashboard, I tried to do this by translating raw data into plain English. Instead of showing a chart of SHAP values, the system generates sentences describing if a factor for example abnormal thalassemia is raising or lowering the risk and by how much. The intention was to make the reasoning accessible without needing the user to understand the mathematics and data.

But the moment you simplify a complex output into a sentence you end up making a choice about what to include, what to leave out and how to say it. These choices are never neutral. A feature described as “strongly increasing risk” carries a different weight than one on a bar chart. Language implies certainty and in medicine, false certainty is dangerous.

Looking ahead: what this means for the future of healthcare

Healthcare AI isn't going anywhere. The question isn't if these systems will be used but under what conditions? with what safeguards? and with whose interests?

For patients, the stakes are immediate. As AI assisted tools become more common in diagnostics and risk assessment, people will find more decisions shaped by algorithms that they have no ability to question. Healthcare is already marked by significant inequalities, and systems trained on unrepresentative data can deepen those divides. The UCI Cleveland dataset used in my project for example, was collected in the 1980s from a single clinical population. A model trained on it reflects the patterns of *that* population and not the diversity of patients in a modern NHS clinic. Without that context, a confident AI prediction can be wrong for some people.

For developers and researchers and for people who build these systems as students, this raises a responsibility that sits with the technical work. In my own project, choosing the UCI Cleveland dataset was a practical decision. But it was also a choice about whose experience the system would reflect most accurately. The choices made during development, which dataset to use, how to handle missing data, which metrics to prioritise, how to phrase an explanation, are not just technical decisions. They embed assumptions about who the system is for, what counts as a good outcome, and whose experience is treated as the default. Recognising this is not just good ethics, it is good engineering.

Regulation is starting to catchup. The EU AI Act, once fully enacted, would place significant obligations on developers of high-risk AI systems in healthcare, requiring transparency, human oversight, and ongoing monitoring. But regulations only set a floor, not a ceiling. The more important shifts are cultural, moving from a world where AI systems are evaluated mainly on benchmark performance to one where usability, fairness and honest communication of uncertainty is treated as first class concerns.

The most important lesson from this project is not a technical one. It is that building a system people trust requires more than building a system that does well. It requires honesty about what the system does not know, and thought about who it might fail, and a commitment to explanation that informs.